

Subject Description Form

Subject Code	ITC3204T
Subject Title	Textile Materials Foundation
Credit Value	3
Level	3
Pre-requisite(s) / <Co-requisite> / (Exclusion)	Nil
Objectives	The subject provides the fundamental knowledge of typical polymers, fibers pertaining to textiles in biological, chemical, physical, mechanical, structural and technological terms.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - understand the fundamentals of polymer chemistry; - understand the structures and properties of different polymers and fibers as textile materials; - explain the relationships among structures, properties and applications to textiles of different materials; - identify key features of different fibers for textiles; - test and evaluate essential properties, structural parameters and performance of typical textile materials; - expound the relationships of key features and technological requirements of various materials for textiles.
Subject Synopsis/ Indicative Syllabus	(I) Polymers (50%) Basic terminology in polymer science, structures and physical properties of polymer typically used in textiles, polymerisation, crystalline state, amorphous state, and structure –property relation. (II) Fibres (50%) Classification, structures, physical and mechanical properties and manufacturing technologies and performance of fibers for textiles.
Teaching/Learning Methodology	Lectures, tutorials and labs will be used for teaching and learning.

Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)					
			a	b	c	d	e	f
	Continuous Assessment	50%	✓	✓	✓	✓	✓	✓
	1. Test	25%	✓	✓	✓	✓		✓
	2. Laboratory report	25%	✓	✓	✓	✓	✓	✓
	Examination	50%	✓	✓	✓	✓	✓	✓
	Total	100%						
	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:							
	This subject will provide theoretical lecturing, tutorial and laboratory hands-on experiences to students on textile materials. Therefore, it will be necessary to carry out coursework to assess the student’s learning outcomes at different stages, particularly in practical aspects of the subject, as well as to have final examination to assess the theoretical understanding of the subject contents.							
Student Study Effort Expected	Class contact:							
	• Lecture						26 Hrs.	
	• Lab						6 Hrs.	
	• Tutorial						6 Hrs.	
	Other student study effort:							
	• Self-study						67 Hrs.	
	Total student study effort						105 Hrs.	
Reading List and References	<u>Books</u>							
	Carraher, C. E. (2016), <i>Carraher’s Polymer Chemistry</i> . 9th edition, CRC Press, US.							
	Eichhorn, S. J. (2009), <i>Handbook of Textile Fibre Structure: Volumes 1 and 2, edited</i> . Woodhead Publishing, Cambridge.							
	Hearle, J., Morton, W. E.. (2008), <i>Physical Properties of Textile Fibres</i> . 4 th Edition, CRC Press, US.							
	Young, R. J., Lovell, P. A. (2011), <i>Introduction to Polymers</i> . 3rd edition, CRC Press, US.							
	Zhang, X. (2014), <i>Fundamentals of Fiber Science</i> . Destech Publications Inc, US.							